End-to-End MLOps Workflow

1. Install Required Packages

```
py -m pip install fastapi uvicorn pandas scikit-learn joblib
```

---

2. Run ML Pipeline Locally

```
python data_ingestion.py

py train.py

py model_training.py

py save_model.py

py predict.py
```

---

3. Run FastAPI Application

```
uvicorn app:app --reload

Open:
http://127.0.0.1:8000/docs
```

---

4. Test Prediction API

```
POST /predict

{
  "CustomerID": 3001,
  "Age": 35,
  "Gender": 1,
  "MonthlyCharges": 5000,
  "Tenure": 24
}
```

Prediction Meaning

```
0 → Customer will NOT churn
1 → Customer WILL churn
```

5. Docker Workflow

```
docker --version

docker build -t churn-api .

docker run -p 8000:8000 churn-api
```

6. Docker Compose

```
docker compose up
```

7. GitHub Actions CI/CD

```
Workflow Location:
D:/ML/.github/workflows/main.yml

Git Push
    ↓
GitHub Actions Trigger
    ↓
Install Dependencies
    ↓
Run ML Training
```

8. Git Commands

```
git init
git add .
git commit -m "Initial MLOps project"
git branch -M main
git remote add origin https://github.com/shekharglobal/mlops-project.git
git push -u origin main
```

9. .gitignore

```
__pycache__/
*.pyc
```

10. Docker Hub Workflow

```
docker login

docker tag churn-api 110044/churn-api:v1

docker push 110044/churn-api:v1

docker pull 110044/churn-api:v1

docker run -p 8000:8000 110044/churn-api:v1
```

---

11. Final Architecture

```
Dataset
   ↓
Data Ingestion
   ↓
Preprocessing
   ↓
Model Training
   ↓
Model Saving
   ↓
Prediction API
   ↓
Docker Container
   ↓
Docker Hub
   ↓
GitHub Actions CI
```

---

12. ML Logic

```
Model learns customer behavior patterns:

High MonthlyCharges + Long Tenure → No Churn
Low MonthlyCharges + Very Short Tenure → Churn

f(X) = Y

X = Customer Features
Y = Churn Prediction
```

---

13. Skills Learned

```
✓ Python ML Pipeline
✓ FastAPI Deployment
✓ Dockerization
✓ Docker Compose
✓ GitHub Integration
```

✓ GitHub Actions CI
✓ Docker Hub Push/Pull
✓ REST API Testing
